

Improving Long-Rollout Accuracy of the DNO Surrogate

Two training-time approaches we are evaluating

Jonathan Ma

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The rollout problem

The surrogate learns to predict the Dirichlet–Neumann operator $\mathcal{G}(\eta)\xi$ from snapshots of (η, ξ, h) , then is iterated forward in time with a Zakharov integrator.

Pointwise (single-step) accuracy is high. Long-time rollouts are not.

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On evaluation regimes (Tanaka solitons, Benjamin–Feir, Stokes, random sea, linear), we observe error that grows *coherently in time*, dominated by modes the model is *supposed* to resolve — not by aliasing, not by high- k noise.

The question: how do we train the surrogate to remain accurate when fed its own predictions?

Why one-step training is insufficient

Standard objective: given a data pair $(\eta, \xi) \mapsto \mathcal{G}(\eta)\xi$, minimise a Sobolev- H^k loss against the analytical target.

What this misses:

- The model only ever sees data-distribution inputs.
- At rollout, it consumes its own prediction as input.
- Small biases at one step compound coherently into the next step's input distribution.

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A loss that is locally accurate is not the same as a dynamical system that is globally stable. The two approaches below directly target this gap.

Approach 1: Pushforward training

Idea. Force the surrogate to be accurate at states *it would itself produce* after a few rollout steps.

Given a data point (η, ξ) , advance the surrogate's own prediction through k Euler steps of the Zakharov system, then re-predict the DNO at the rolled-out state:

$$(\eta_k, \xi_k) = \text{Step}^k(\eta, \xi; \mathcal{G}_{\text{pred}}), \quad \text{pred}_k = \mathcal{G}_{\theta}(\text{sg}(\eta_k), \text{sg}(\xi_k)).$$

The loss adds a term comparing pred_k against the analytical Craig–Sulem expansion at the rolled-out state:

$$\mathcal{L} = \mathcal{L}_{\text{single-step}} + \lambda \mathcal{L}(\text{pred}_k, \mathcal{G}_{\text{CS}}(\eta_k, \xi_k)).$$

Gradients flow only through the final prediction: the trajectory and the target are treated as fixed.

Approach 1: Pushforward — intuition and limits

What it teaches. The surrogate must be accurate not just on the training manifold, but on the (slightly off-manifold) states it itself generates. This breaks the covariate-shift compounding explicitly.

What it does not address.

- No notion of physical conservation laws.
- Small k ($k=1$) may not capture long-horizon coupling; large k is expensive and harder to optimize.
- Stop-gradient on the rolled-out state introduces a known bias scaling like k^2 relative to the true multi-step gradient (List, Thuerey et al., 2024).

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Status. Currently trained at $k=1$, $\lambda=1$. A longer-horizon, gradient-through unroll is a planned next step.

Approach 2: Hamiltonian-conservation loss

Idea. The Zakharov system is Hamiltonian, with

$$H(\eta, \xi) = \frac{1}{2} \int_{\Omega} (\xi \mathcal{G}(\eta) \xi + g \eta^2) dx.$$

An exact solution conserves H . A surrogate whose rollout drifts in H is producing unphysical energy creation or destruction — even if its pointwise error per step is small.

Add a loss term that penalises one-step relative drift of H :

$$\mathcal{L}_H = \lambda_H \mathbb{E} \left[\min \left(\frac{|H_1 - H_0|}{|H_0| + \varepsilon}, C \right) \right],$$

with H_0 evaluated at the data state and H_1 at one Euler step of the surrogate's own dynamics. Gradient flows through both $\mathcal{G}$ -predictions; the integrator is treated as fixed.

Approach 2: H-cons — intuition and limits

What it teaches. A physically-motivated soft constraint: “predictions that violate energy conservation are penalised, in proportion to how much they violate it.”

What it does not address.

- Single-step horizon only — does not directly stabilise long rollouts the way pushforward does.
- H depends on $\mathcal{G}$, which is what we are learning — so the loss is self-consistency of the surrogate’s energy belief, not conservation of the true energy.
- Does not enforce momentum, mass, or other invariants.

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Status. Currently training with linear warmup and per-sample clip to bound early-training transients. First evaluation pending.

Pushforward, vs. baseline at matched compute (partial).

- Median relative L_2 error at the final time horizon is reduced by roughly 30–70% on Tanaka, Benjamin–Feir, and the linear regime.
- Stokes regimes are essentially unchanged (error already small).
- Random sea regimes regress — pushforward at $k=1$ appears to trade noise-robustness on broadband ICs for fidelity on coherent, soliton-like ICs.

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Caveats.

- Only 16 ICs per regime — per-regime comparisons have wide confidence intervals.
- Pushforward run is still training to matched epoch count.
- H-cons evaluation not yet available.

What we are working on next

- Finish the pushforward run to matched compute against the baseline.
- Evaluate the H-conservation run on the same suite.
- Investigate the random-sea regression: is it a pushforward-horizon issue, or does it need a different IC sampling distribution during training?
- Larger k , gradient-through unrolls (list, Thuerey 2024 suggests $k \geq 8$ is where this becomes stable).
- Tendency-target reformulation (predict $\partial_t u$, then integrate with RK4 externally)
— decouples the surrogate from the dispersive phase rotation.